

VG SERIES 3 – EPISODE 5 : Between Mass and Spin – On Inertia, Rotation, and the Two Faces of the Hamiltonian

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Abstract

This episode clarifies once and for all the relationship between the two forms of the Hamiltonian of the Primary Network: the simplified form (the one-inch equation) and the extended form (with spin terms). It explains why effective mass was introduced and later withdrawn, showing that spin is more fundamental than mass, and torsion is more fundamental than gravity. The episode includes a chapter dedicated to black holes as laboratories of transition — where the two forms of the Hamiltonian find their natural application. It is an invitation to view the Network not as a machine, but as a dance of rotations.

Keywords: spin, effective mass, torsion field, Hamiltonian, one-inch equation, black holes, regime transition, VG Series 3.

1. INTRODUCTION – WHY WE NEED THIS CLARIFICATION

As RTS has grown, several mathematical tools have gathered around it. Some have become fundamental, others have remained in the background, and others have been introduced and then withdrawn. This is natural — a living theory grows through attempts, not through dogmas.

This episode is not a correction of previous ones. It is a clarification — a pause to look back and say: „This is what we did, why we did it, and how it helps us now.”

2. THE TWO FORMS OF THE HAMILTONIAN

The simplified form – the one-inch equation

This is the backbone of RTS. It was introduced in RTS16 and used in all previous episodes, in the form:

$$H = \sum_i \left(\frac{p_i^2}{2} + \frac{1}{2} \omega_i^2 q_i^2 \right) + \sum_{i < j} K_{ij} q_i q_j$$

In VG2 (Series 3), an extension with effective mass terms was introduced to describe the inertia of nodes in extreme regimes. Although this extension was later withdrawn, it remains an important example of theoretical development.

The extended form – with spin terms

In VG2 (Series 3), this equation was extended to explicitly include spin:

$$H = \sum_i \left(\frac{p_i^2}{2} + \frac{1}{2} \omega_i^2 q_i^2 \right) + \sum_{i < j} K_{ij} q_i q_j + \sum_{i < j} \lambda_{ij} (\hat{S}_i \cdot \hat{S}_j) q_i q_j + \sum_{i < j} \gamma_{ij} (\hat{S}_i \times \hat{S}_j) \cdot q_i p_j$$

These two terms add:

- **Spin-spin coupling** — how spin alignment influences interaction.
- **Spin-orbit coupling** — rotation introduced into the Network's dynamics.

3. WHY WE INTRODUCED MASS AND WHY WE WITHDREW IT

Reason for introduction

Effective mass was introduced to describe the inertia of nodes in extreme regimes — where couplings are so intense that nodes seem to resist reconfiguration. Mass was a way of saying: „This node moves more slowly.”

Reason for withdrawal

It was withdrawn because:

- Mass is not fundamental — it is an effect of nodal couplings.
- Introducing it unnecessarily complicated the equation for most applications.
- It was decided to keep simplicity where possible, and to reserve extensions for cases that truly require them.

What was learned from this attempt?

- Mass is an echo — not a source.
- Spin is the source — everything begins with rotation.
- Torsion is the voice of spin — the way the Network tells its own story.

4. SPIN – THE OLDEST DANCE OF THE NETWORK

Spin is the oldest dance of the Network. Before matter existed, before gravity curved space, before time flowed, spins were already rotating. They are the source of torsion, and torsion is the source of gravity. What we call mass is nothing but an echo of this dance — a shadow cast by the rotation of nodes into the macroscopic world.

How exactly is spin the source of torsion?

To understand this relationship, we must look at the **singularity** — not as a point of infinite density, but as a **critical nodal density threshold**. At this threshold, the couplings between nodes become so strong that the geometry of spacetime ceases to make sense, and the **spins of the nodes are forced to align**. This forced alignment of intrinsic rotations generates a **coherent field of twist** — the **torsion field (TF)**. In turn, this torsion field becomes the **source of gravity** and of the other forces.

Thus, in RTS, the sequence is:

spin → torsion → gravity

What does this mean for RTS?

- Spin is the root — everything starts from it.
- Mass and gravity are only side effects of a much older dance.
- The torsion field is the voice of this dance — the way the Network tells its own story

What does this mean for applications?

When studying black holes or phase transitions, we should not look for mass or gravity as separate entities. We should look at the dance of spins — how they align, reorient, and reverse flux.

5. WHEN TO USE EACH FORM

- **The simplified form** is sufficient for:
 - General cosmology.
 - Macroscopic phenomena.
 - Weak interactions.
 - Most theoretical applications.
- **The extended form** is necessary for:
 - Black holes and phase transitions.
 - Torsion fields.
 - Extreme spin phenomena.
 - Any situation where the rotation of nodes plays an essential role.

6. BLACK HOLES AS LABORATORIES OF TRANSITION

Black holes are the places where the two forms of the Hamiltonian find their natural application. They are laboratories of transition — where regimes shift, spins reorient, and absorption becomes emission.

Porphyron — a cosmic jet 23 million light-years long

This giant jet is an example of a regime transition — a black hole temporarily behaving like a white hole, ejecting matter and energy at cosmic scales, without giving birth to a daughter Network. It is a flux reversal, not a birth.

J2245+3743 — the eruption of 10 trillion suns

This violent eruption is another example of flux reversal — a black hole that temporarily entered an emission regime, releasing a colossal amount of energy.

What do these cases teach us?

- Regime transition is a real, observable phenomenon.
- The birth of a daughter Network is a rarer event, requiring extreme conditions.
- The distinction between the two is essential for understanding Network dynamics.

What is permanent in the universe?

In RTS, only the following are permanent:

- The constant number of nodes (NN) — nodes are indivisible and eternal.
- The Network itself — as a fundamental entity.
- The basic laws of couplings — which govern all transformations.

Everything else is temporary:

- Nodal configurations (particles, atoms, stars, galaxies).
- Absorption and emission regimes.
- Even the cosmic cycles of expansion and contraction.

A white hole emits until it exhausts itself, then becomes a shell. A black hole absorbs until it evaporates, then leaves behind the same shell. In the end, both — regardless of their history — turn into inert shells, which through coalescence contribute to dark matter. There are no permanent WH or BH in RTS. What is permanent is the Network itself.

7. CONCLUSION – A SYNTHESIS THAT OPENS TOWARD THE FUTURE

VG5 is not an end. It is a synthesis — a place where the threads of the theory have been gathered and placed into a coherent image. We have shown that:

- Spin is more fundamental than mass.
- Torsion is more fundamental than gravity.
- The two forms of the Hamiltonian are complementary, not competing.
- Black holes are the natural laboratories of this distinction.

This clarification is now written. It can be read, discussed, adjusted. But it exists.